Commodities

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Contents

1 Market Foundation

Core Concept 1.1: Commodity Markets

Commodity markets deal primarily in **physical, tangible assets** that serve as **elementary inputs** for global economic production. Because these raw materials form the base of the manufacturing supply chain, significant price volatility or supply disruptions in these goods inevitably cascade down to consumer-end products. *For instance, while wheat represents the underlying commodity traded in financial markets, commercial bread serves as the final consumer good.*

Modern tradable commodities are systematically classified into three primary sectors.

- **Energy** (*Section 4*): Primary fuel resources, including crude oil, natural gas, and coal.
- **Metals** (*Section 5*): Industrial and mineral ores, spanning base, ferrous, and precious varieties.
- **Agriculture** (*Section 6*): Cultivated assets, comprising grains, oilseeds, soft commodities, and livestock.

Core Concept 1.2: Historical Development of Commodity Trading

Commodity trading represents one of the oldest forms of economic commerce, predating the development of equity and fixed-income markets by several millennia, with historical origins extending back to ancient Mesopotamia.

Early transactions functioned as forward negotiations between parties to exchange a specific quantity of a commodity at a designated future date—a mechanism known today as an **over-the-counter (OTC)** contract. However, persistent disagreements regarding quality standards and delivery terms eventually drove the market toward standardization, giving birth to centralized futures trading. Early global milestones in this institutional evolution include the **Dojima Rice Exchange in Japan** (*1697*), the **Chicago Board of Trade (CBOT)** (*1848*) in the United States, and, subsequently the **London Metal Exchange (LME)** (*1877*) in Europe.

Historically, these centralized marketplaces relied on a physical **open-outcry** system (*Figure 1*), a framework where floor brokers executed transactions face-to-face within trading pits through synchronized shouting and hand signals.

Figure 1: Chicago Board of Trade open-outcry system termed ”*the Pit*”

In the modern era, however, commodity execution has almost entirely transitioned to electronic platforms. As a result of this widespread digital shift, the LME remains unique as one of the final venues to preserve its legacy open-outcry system termed ”*the Ring*”.

Core Concept 1.3: Basic Supply & Demand

Commodity markets are governed by the basic economic principles of **supply and demand** (*Section 2*). Within this framework, **physical storage** (*Figure 2*) serves a critical stabilizing function, dampening price volatility and restoring market equilibrium during imbalances, such as an unexpected supply shock.

Figure 2: Japanese oil reserves; *in March 2026, Japan announced the relase of about 80 million barrels in response to the Middle East conflict in Iran, which has resulted in the disruption of oil flows through the **Strait of Hormuz***

Because production centers are rarely geographically aligned with primary consumption hubs, logistics play a defining role in market dynamics. Over 90% of global commodity trade volume is transported via **maritime freight** (*Figure 3*), with the remainder distributed across pipelines, rail, trucking, and aviation—meaning that transportation costs represent a significant economic variable that heavily influences both trade routes and ultimate market destinations.

Figure 3: Oil tanker

Accurate analysis of these dynamics, however, is complicated by significant **data acquisition challenges**: sovereign nations report production, consumption, and inventory figures with varying degrees of frequency, scope, and statistical reliability, requiring market analysts to dedicate substantial effort to modeling and estimating these baseline values.

2 Commodity Trading Mechanics

2.1 Market Structure & Price

Core Concept 2.1: Physical Pricing and OTC Structure

Commodity valuation requires the **simultaneous evaluation of multiple physical variables**, where variations in quality, grade, and geographic location heavily influence spot pricing.

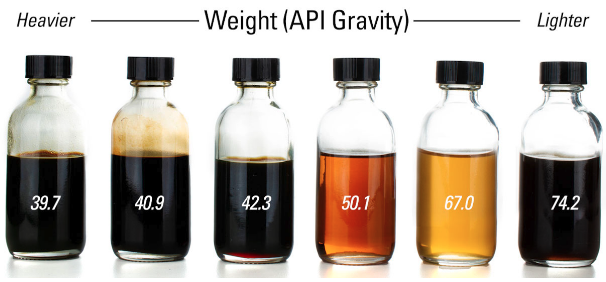

Figure 4: Basic crude grade spectrum

In the oil sector, for example, **hundreds of distinct crude grades** are traded globally, meaning the limited selection of standardized futures contracts **cannot always accommodate the precise needs of physical market participants**. Consequently, physical commodity commerce remains predominantly structured as a **bilateral business**, directly connecting buyers and sellers via customized OTC transactions. *This contrasts with exchange-based markets, where standardized contracts enable multilateral participation among a large number of buyers and sellers but offer substantially less flexibility in contract design.*

Core Concept 2.2: Reference Pricing and Benchmarking

To establish reliable baseline values amid the complexities outlined in Core Concept 2.1, the market relies on specialized **Price Reporting Agencies (PRAs)** such as **S&P Global Platts** or **Argus Media**, which publish independent price assessments across a broad spectrum of contracts. These assessments provide critical transparency into the market, allowing the majority of physical OTC transactions to be executed at an agreed price differential **relative to these recognized benchmarks**.

Furthermore, because **standardized financial futures do not exist for every unique physical grade**, market participants rely on these benchmarks as pricing proxies to manage asset exposure. In practice, a physical OTC agreement for a niche commodity is structured around a direct peg to a dominant benchmark, adjusted by an agreed-upon **quality** or **location spread**.

Once the physical transaction is linked to the benchmark, the producer or consumer can execute an equal and opposite position in the financial futures market. This strategy creates a dual-market balancing mechanism: any subsequent loss in physical revenue caused by fluctuating spot prices is offset by a corresponding gain in the futures account, and vice versa. Ultimately, while this mechanism requires participants to **forgo the potential upside of favorable price movements**, it **secures operational certainty by locking in baseline profit margins**.

(a) **Producer Hedging Against Brent Price Fall**

A producer selling a discounted crude grade can contractually peg their sales price to `Brent - $3.00/bbl` while simultaneously **shorting** Brent financial futures at $75.00. If the market declines, the reduced revenue from the physical delivery at $62.00 is precisely counterbalanced by a $10.00 gain in the short position, thereby securing the targeted net effective return of $72.00 per barrel.

(b) **Refiner Hedging Against Brent Price Rise**

Conversely, a refiner purchasing a discounted crude grade can contractually peg their physical acquisition price to `Brent - $3.00/bbl` while simultaneously executing a **long position** in Brent financial futures at $75.00. If the market rallies, the increased procurement cost from the physical delivery at $82.00 is precisely counterbalanced by a $10.00 gain in the long position, thereby securing the targeted net effective purchase cost of $72.00 per barrel.

Attribute	Producer Hedge	Refiner Hedge
Objective	Protect against falling Brent price	Protect against rising Brent price
Contract	Sell at `Brent - $3.00/bbl`	Buy at `Brent - $3.00/bbl`
Futures Position	Short Brent futures at $75.00/bbl	Long Brent futures at $75.00/bbl
Market	Price declines to $65.00/bbl	Price rallies to $85.00/bbl
Delivery Price	$62.00/bbl	$82.00/bbl
Futures Gain	+$10.00/bbl (*short profit*)	+$10.00/bbl (*long profit*)
Net Effective Price	$72.00/bbl	$72.00/bbl

Table 1: Hedging scenarios for brent-linked crude pricing

Core Concept 2.3: Regional Markets and Arbitrage Dynamics

Crude oil does not trade at a singular, unified global price, operating instead through decentralized regional markets that rely on **localized benchmarks**. Under standard market conditions, the price differentials between these geographic boundaries remain fundamentally constrained; whenever localized imbalances widen these spreads beyond baseline transport costs, market participants exploit the resulting **arbitrage**, shifting physical supply and demand pressures until **prices realign**.

Core Concept 2.4: Investment Drivers and Market Financialization

As commodity markets have matured and gained liquidity, they have attracted a growing volume of purely financial participants who do not directly contribute to the physical value chain. This capital influx and participation in commodities markets is motivated by the following set of investment objectives.

- **Capital preservation** as a safe haven during periods of broader macroeconomic uncertainty, with tangible assets such as gold serving as a classic instrument for this purpose
- **Hedging** against macroeconomic uncertainty through tangible commodities to preserve capital.
- **Portfolio diversification** through exposure to asset classes exhibiting historically low correlation with traditional securities such as equities and fixed income
- **Speculative positioning** (*Core Concept 2.7*) to capture profits from the inherent price volatility of commodities

Like traditional financial assets, commodity spot prices are heavily responsive to shifts in broader market sentiment. This behavioral dynamic has become significantly more pronounced alongside the progressive **financialization of commodity markets** (*Subsection 2.2*), as maturing market liquidity has attracted a growing volume of purely financial participants who do not directly contribute to the physical value chain yet actively trade commodity contracts.

Core Concept 2.5: Price Drivers

The market valuation of primary commodities is dictated by a diverse array of structural, environmental, and institutional drivers.

(a) **Weather & Natural Disasters**

Environmental conditions and severe weather anomalies exert direct, immediate shocks on regional commodity supply and demand.

For instance, an unseasonal cold wave in the United Kingdom can rapidly escalate residential heating requirements, triggering a sharp increase in regional natural gas prices due to the sudden demand surge.

(b) **Geopolitical Events**

Because no sovereign nation is entirely self-sufficient, global economic stability depends heavily on an interconnected network of commodity imports and exports. Consequently, any significant disruption to supply or demand dynamics within key trading nations can precipitate adverse cascading effects throughout the global economy.

Figure 5: Crude oil price response to escalating tensions associated with the U.S.-Isreal-Iran conflict in March 2026

This is perfectly exemplified by the surge in crude oil prices stemming from the 2026 Iran-US-Isreal conflict (Figure 5).

(c) **Policy & Regulation**

Regulatory mandates and institutional policies frequently reshape demand configurations, directly affecting the profitability of refining and processing firms based on their specific technological and equipment constraints.

A prominent example is the International Maritime Organization (IMO) regulation mandating that all marine vessels utilize fuel with a sulfur content of 0.5% or less, which rapidly altered refinery margins globally by shifting demand toward low-sulfur distillates.

(d) **Demand Elasticity**

Given that primary commodities function as essential societal **necessities**, aggregate consumer demand tends to be **relatively price-inelastic in the short term**, meaning that broader economic slowdowns typically manifest as **gradual, lagging shifts** rather than immediate disruptions.

There are, however, extreme exceptions to this trend. The outbreak of the **COVID-19 pandemic** rapidly decimated nearly one-third of global fuel demand. This unprecedented collapse in near-term consumption (*Figure 7*) forced the crude oil market into an extreme **super contango**, where immediate physical spot prices traded at a dramatic discount relative to longer-dated futures contracts.

2.2 Financialization of Commodities

Core Concept 2.6: Historical Evolution

While physical supply and demand fundamentals remain foundational drivers of commodity valuation, broader market sentiment also exerts a significant influence on asset prices. This dynamic has intensified over the past few decades due to the progressive **financialization of the global oil market**, characterized by a substantial influx of non-traditional market participants actively trading energy derivatives.

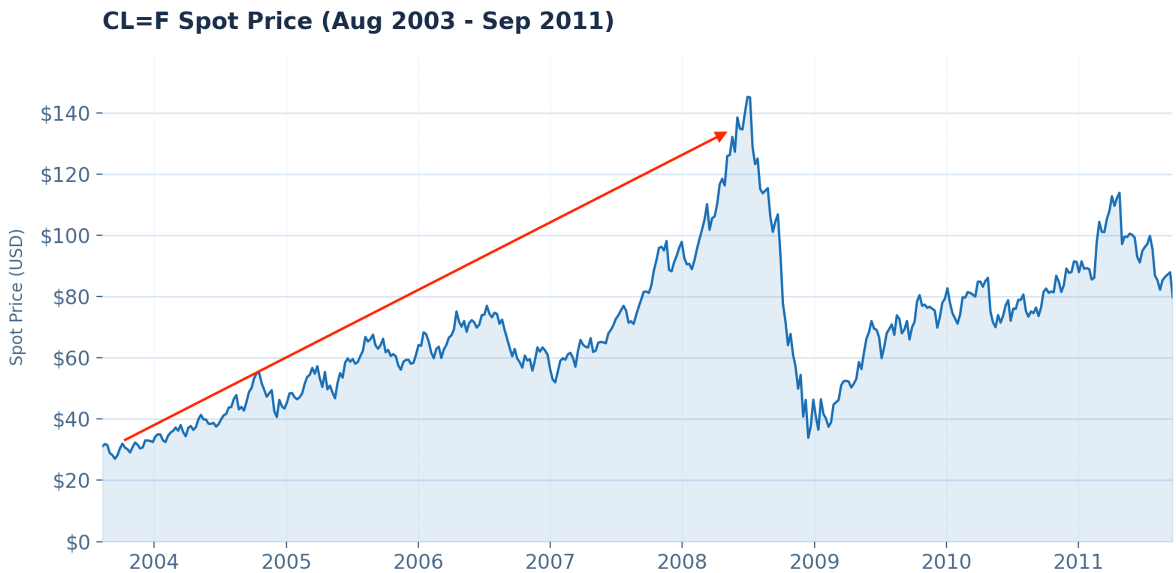

Figure 6: Crude oil future spot from Aug 2003 to Sep 2011

A prominent economic perspective suggests that the dramatic surge in oil prices between 2003 and 2008 (*Figure 6*) cannot be fully accounted for by physical supply and demand dynamics alone. Instead, this appreciation was heavily facilitated by the rapid financialization of oil futures markets—where trading volumes multiplied by 20 to 30 times and capital allocations expanded from less than $10 billion to $450 billion by April 2011—effectively transitioning these physical asset marketplaces into highly financialized arenas.

Core Concept 2.7: Speculation & May 2020 WTI Contract

Speculation involves the acquisition of assets at lower valuation baselines with the intent of selling at higher prices in the future. However, when aggressive speculative capital clashes with severe physical constraints, extreme structural market failures can occur. This vulnerability was vividly exposed during the early stages of the COVID-19 pandemic, when unprecedented global lockdowns triggered an immediate, massive collapse in oil consumption while physical crude production remained highly inelastic, forcing **millions of surplus barrels** into localized distribution networks.

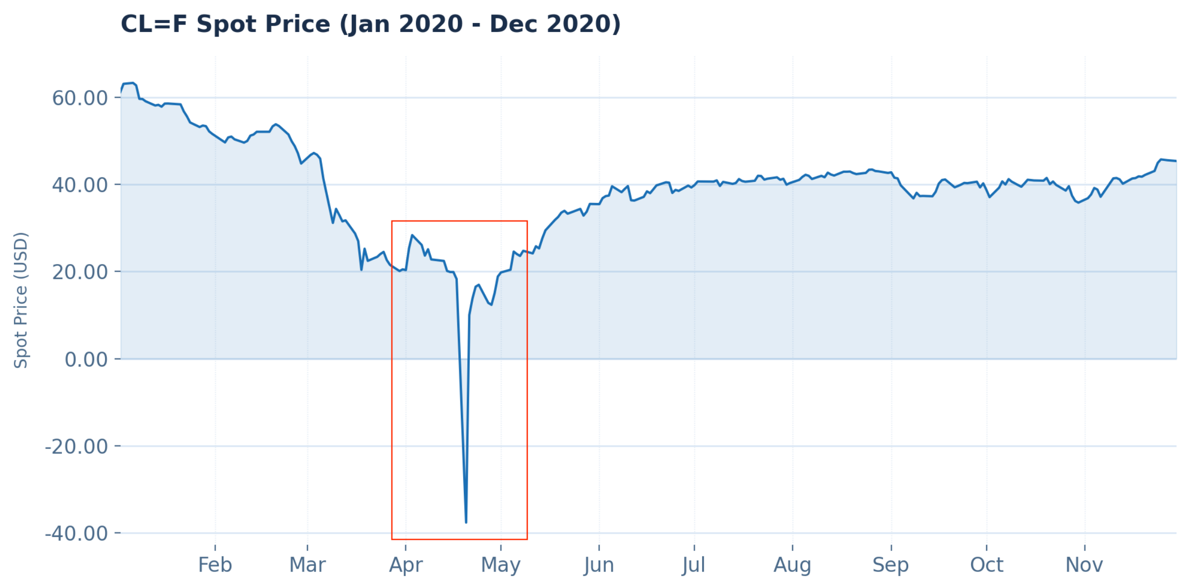

Figure 7: WTI front-month futures prices from January 2020 to December 2020

As illustrated in Figure 7, this supply-demand mismatch culminated in the historic **price collapse of the May 2020 WTI contract**. In the weeks preceding the crash, retail and institutional speculators aggressively channeled capital into crude oil ETFs and passive index products, attempting to capture the market bottom in anticipation of an economic rebound. Crucially, these financial participants overlooked the operational mechanics of the WTI contract, which mandates compulsory physical delivery upon expiration at Cushing, Oklahoma—the primary domestic pipeline crossroads with a working storage capacity of approximately 90 million barrels.

On the eve of expiration, this massive influx of speculative capital encountered an absolute physical bottleneck. Because the global supply glut had filled Cushing's storage facilities to maximum capacity, non-physical market participants holding long positions faced a structural crisis. Lacking the infrastructure to accept physical delivery, these traders were legally required to roll their expiring May contracts into June by liquidating them at any cost. With a massive wall of forced sellers and zero available storage buyers, the market suffered a complete technical breakdown, ultimately driving prices into negative territory for the first time in history.

3 Commodity Players

3.1 Producers & Consumers

Core Concept 3.1: Producers & Consumers

Market participants within the commodity ecosystem are broadly classified into producers and consumers. **Producers** encompass upstream entities such as oil extraction firms and agricultural operations, whereas **consumers** include industrial manufacturing corporations and aviation firms that require physical raw materials for production.

The strategic objectives of these two groups are inherently opposing: producers seek to secure the highest possible selling prices to protect revenues, while consumers aim to lock in the lowest feasible purchase prices to insulate margins. *The aviation sector heavily relies on price locking via futures, as jet fuel procurement frequently accounts for 20% to 50% of total corporate operating costs.*

3.2 Trading Houses

Core Concept 3.2: Role as Intermediaries & Vertical Integration

Direct transactions between primary producers and end consumers are relatively rare, with specialized **trading houses** typically intermediating these supply chains to manage the underlying logistics and import operations, alleviating the administrative and operational burdens on consuming firms.

Rather than operating strictly as passive intermediaries, however, **modern trading houses frequently integrate vertically** by directly engaging in the extraction, refining, and processing of raw materials. Additionally, they leverage extensive logistical networks to source physical assets from a diversified global supplier base and distribute them efficiently to international end markets.

Core Concept 3.3: Arbitrage Strategies

Trading houses primarily capture profit margins by identifying and exploiting structural **market inefficiencies** through three principal **arbitrage** strategies.

(a) **Geographical arbitrage** involves purchasing a physical commodity at a lower price in one market and absorbing the freight costs to sell it at a premium in another location.

(b) **Temporal arbitrage** entails acquiring inventory during periods of localized oversupply or low seasonal demand, managing its storage, and liquidating the asset when prices appreciate in the future.

(c) **Production arbitrage** leverages processing capabilities, allowing firms to procure lower-cost raw materials and transform them into higher-value refined commodities.

Core Concept 3.4: Major Players

As of June 2026, the global commodity intermediation landscape is highly consolidated, dominated by two principal groups of multinational trading conglomerates.

Figure 8: Glencore mining operations

- **Energy & Metals Merchants**: Glencore (`GLEN`), Vitol (*private*), Trafigura (*private*), Gunvor (*private*), Mercuria (*private*), Koch (*private*)
- **Agricultural Conglomerates**—collectively known as the **"ABCD"** participants: Archer-Daniels-Midland (`ADM`), Bunge (`BG`), Cargill (*private*), Louis Dreyfus Company (*private*)

3.3 Investment Banks

Core Concept 3.5: Investment Banks

Investment banks fulfill critical financial roles across the commodity value chain by providing essential capital and risk management solutions, such as structuing project financing, raising equity and debt capital, offering working capital solutions, and hedging physical exposure.

Historically, many investment banks actively engaged in physical merchant trading and asset ownership. However, following the **2008 financial crisis**, regulatory reforms and legislative prohibitions compelled most Western institutions to divest from physical operations, though select banks in jurisdictions like Japan and Australia retain these physical capabilities.

3.4 Speculators & Investors

Core Concept 3.6: Speculator vs Investor

Feature	**Speculator**	**Investor**
Time Horizon	Shorter	Longer
Trade Frequency	Trade more	Trade less
Risk Appetite	Higher	Lower
Instruments	Futures, Options, Structured Products	ETFs, Mutual Funds

Table 2: Key distinctions between speculators and investors

While both speculators and investors inject necessary capital into the market, they are distinguished by fundamentally different operational profiles as shown in Table 2. Speculators typically maintain short-term transactional horizons with high trading frequencies and exhibit a high risk appetite, primarily utilizing derivative instruments such as futures, options, and structured products. In contrast, investors adopt long-term capital allocation strategies with low transaction frequencies and a more conservative risk tolerance, generally gaining exposure via collective investment vehicles like ETFs and mutual funds.

4 Energy Markets

4.1 Crude Oil Fundamentals

Core Concept 4.1: Brief History of Oil

Oil has been utilised at a large scale since approximately 6000 BC, with its commercial significance growing substantially following Abraham Gesner's invention of **kerosene** in 1852. Prior to the Industrial Revolution, agricultural commodities constituted the dominant class of traded goods; today, that position is held by **crude oil** (*Core Concept 4.2*). The scope of oil consumption is broad, spanning transportation fuels, industrial inputs, and the manufacturing of everyday household products.

Core Concept 4.2: Crude Oil & Refineries

Crude oil is oil extracted from the ground in its unprocessed state that must undergo refining before it can be used commercially. The refinery process converts raw crude into a range of usable petroleum products, including diesel, jet fuel, and gasoline. Two primary physical characteristics are used to classify crude oil.

- **API Gravity**: API gravity measures crude oil density relative to water, as defined by the **American Petroleum Institute (API)**. Higher API gravity indicates a lighter crude, which is generally more desirable and commands a price premium.
- **Sulfur Content**: Crude oils with lower sulfur concentrations—termed **sweet** crude—are easier and less costly to refine than high-sulfur varieties—termed **sour**.

Refineries differ in both the specifc mix or ”recipe” of different crude oils they are capable of processing—collevtively referred to as the **crude slate**—and the range of end products they can produce. More sophisticated refinery configurations can extract a greater number of refined products from a given barrel of crude. The profit generated from this conversion process is known as the **refinery margin**.

Beyond the refining process itself, fundamentals such as storage capacity, transportation infrastructure, and quality differentials provide critical insight into the supply and demand structure of petroleum markets. Long-term demand for oil is forecast to grow, driven primarily by rising consumption in developing economies.

Core Concept 4.3: Crude Oil Types

Brent West Texas Intermediate (WTI) Dubai / Oman

4.2 Supply Dynamics

Core Concept 4.4: Seven Sisters & Oil Majors

From the 1940s through to the 1970s, a group of seven American and British oil companies—collectively known as the **Seven Sisters**—exercised near-total dominance over global petroleum reserves, controlling approximately 85% of the world’s supply through vertical integration and cartel-like coordination.

Figure 9: The Seven Sisters; *Esso, Mobil, Chevron, Texaco, Gulf, Shell, British Petroleum/BP*

Their monopolistic structure began to erode during the 1970s, driven by two concurrent forces

- The rise of **OPEC** (*Core Concept 4.5*) as an intergovernmental counterweight.
- The assertion of national sovereignty over natural resources by oil-producing nations.

Of the Seven Sisters, four companies survive today—BP, Chevron, ExxonMobil (*Esso*), and Royal Dutch Shell—and are collectively known as the **oil majors**, and the subsequent vacuum was filled by a set of **state-backed national oil companies**, often referred to as the **New Seven Sisters**.

Figure 10: New Seven Sisters; *Saudi Aramco (Saudi Arabia), Gazprom (Russia), CNPC (China), NIOC (Iran), PDVSA (Venezuela), Petrobras (Brazil), Petronas (Malaysia)*

Core Concept 4.5: Organization of the Petroleum Exporting Countries (OPEC)

The **Organization of the Petroleum Exporting Countries (OPEC)**—an initial coalition of Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela—was established in September 14, 1960 during the **Baghdad conference** in Iraq with the primary objective of safeguarding soverign revenues during the monopolistic reign of the Seven Sisters.

Figure 11: Baghdad conference on September 14, 1960

In its early years, OPEC members sought to emulate the pricing practices of the oil majors by setting prices directly; this approach was subsequently replaced by market-based mechanisms, whereby member states influence prices indirectly through the adjustment of production levels.

In the decades following its establishment, OPEC expanded its membership in response to key global events. These developments are discussed in detail in Subsection 4.3.

Core Concept 4.6: Strategic Petroleum Reserves

At the height of its influence, OPEC controlled approximately half of global crude oil production and held roughly 80% of the world's proven reserves. Given the characteristically low price elasticity of demand for oil, this concentration of supply afforded the organisation a substantial degree of market power.

In response to the economic and geopolitical risks posed by this dependency, many consumer nations established **Strategic Petroleum Reserves (SPRs)**—government-controlled stockpiles of crude oil maintained to provide a buffer against supply disruptions and to safeguard national energy security. *Japan's SPR was shown in Figure 2.*

4.3 Historical Price Shocks and Events

4.3.1 The 1970s: Embargoes and Revolutions

Core Concept 4.7: Arab Oil Embargo (1973)

In 1973, Arab oil-producing nations initiated an **export embargo** against the United States and its allies in retaliation for their military support of Israel during the **Yom Kippur War**. Concurrently, OPEC implemented **broader production cuts**, which caused **global oil prices to quadruple** rapidly.

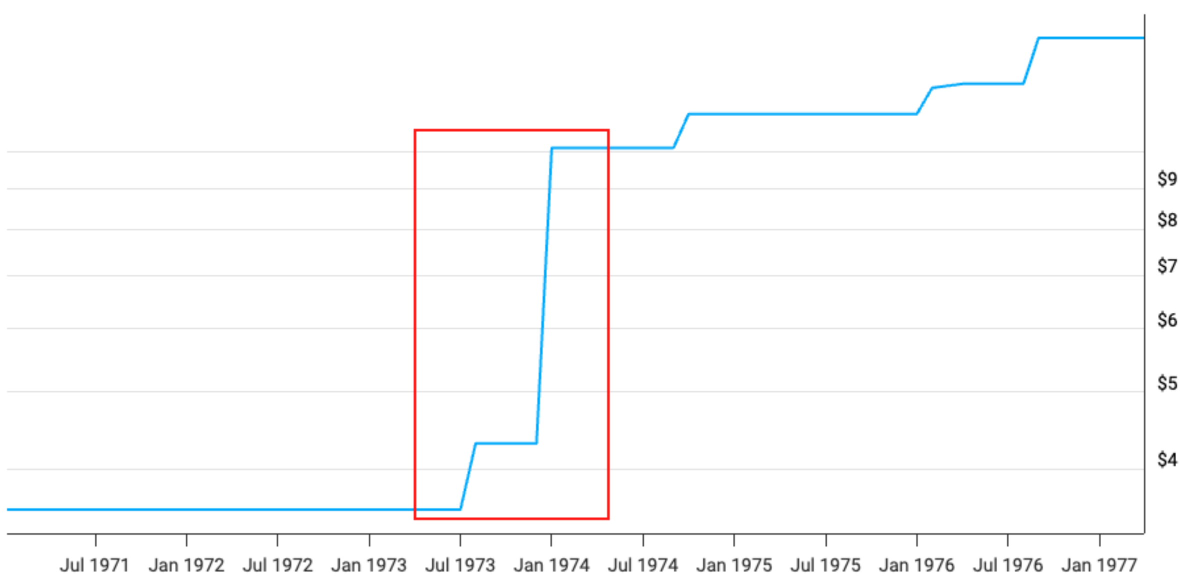

Figure 12: Crude oil prices from Feb 1971 to Mar 1977; *1973 price spike due to the Arab oil embargo and broader production cuts*

While this unprecedented price surge initially generated massive revenue to OPEC member states, it catalyzed profound structural shifts that **ultimately led to severe demand destruction**. The price shock incentivized immediate countermeasures across major economies.

This regulatory shift was mirrored in consumer and industry behavior, evidenced by a marked transition away from inefficient automobiles toward smaller, more fuel-efficient alternatives. Furthermore, the elevated price environment suddenly made capital-intensive exploration in non-OPEC regions—such as the North Sea and Alaska—economically viable, introducing substantial new competitive supply to the market. Confronted by this suppression of global oil consumption, OPEC adjusted its strategic approach, shifting its primary focus from maximizing short-term prices to maintaining long-term market stability.

Core Concept 4.8: Iranian Revolution & The Volcker Shock (1979)

The Iranian Revolution in 1979 caused a severe disruption to the global oil supply as Iran's production and exports plummeted. This unexpected supply shock triggered a panic in the oil markets, leading to a rapid doubling of prices and significantly exacerbating the energy crisis of the 1970s.

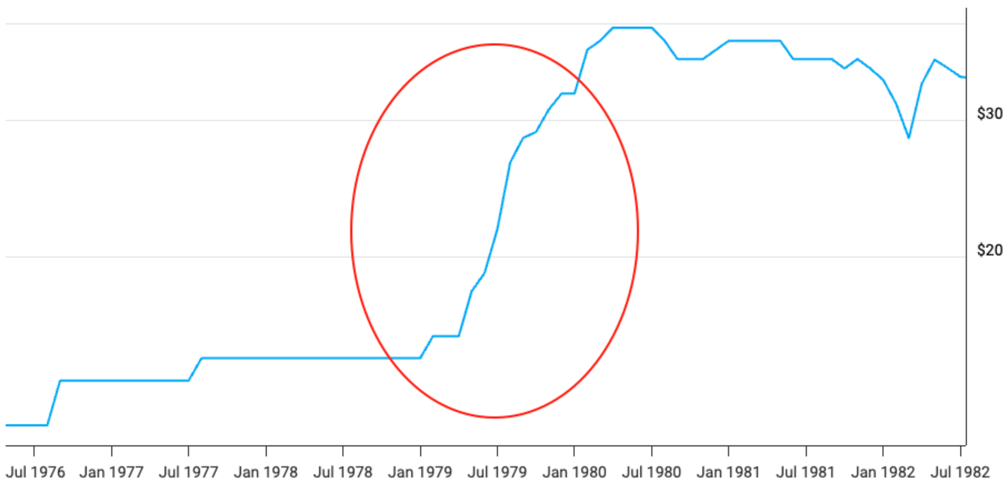

Figure 13: Crude oil prices from May 1976 to Jul 1982; *1979 price spike due to Iranian revolution*

The macroeconomic consequences were severe, pushing the global economy into a period of deep **stagflation**—a damaging combination of high inflation and stagnant economic growth. In the U.S., inflation reached double digits, prompting Federal Reserve Chairman Paul Volcker to intervene drastically in what became known as the **Volcker Shock**. By aggressively raising interest rates to unprecedented levels (*peaking near 20 percent in 1981 as shown in Figure 14*), Volcker successfully halted runaway inflation, but at the cost of triggering a harsh global recession. This resulting economic downturn ultimately crushed global demand for crude oil, setting the stage for the massive **price slump of the 1980s** (*Core Concept 4.10*).

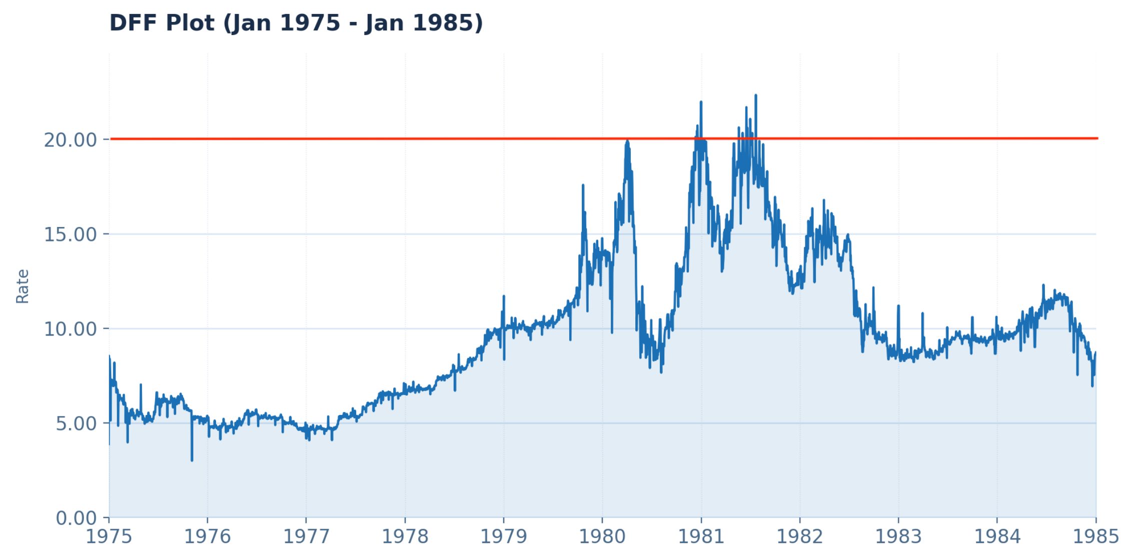

Figure 14: Federal funds rate from 1975 to 1984; *roughly 20% interest rates set by Federal Reserve Chairman Paul Volcker in response to the stagflation triggered by negative oil supply shock due to the Iranian Revoluation of 1979*

4.3.2 The 1980s: Conflict and the Oil Glut

Core Concept 4.9: Iran-Iraq War (1980)

In September 1980, Iraq invaded Iran in an attempt to capitalize on post-revolutionary instability and establish regional dominance. As both nations sought to cripple each other's war-funding capabilities, they systematically targeted opposing oil infrastructure, culminating in the "Tanker War" phase in the Persian Gulf. This mutual destruction initially removed millions of barrels of daily production capacity from the global market

Figure 15: Iran-Iraq war

However, despite the massive scale of the conflict, the anticipated long-term oil shortage never materialized. The demand destruction initiated by the Volcker Shock (*Core Concept 4.8*), coupled with aggressive production increases from Saudi Arabia and emerging non-OPEC suppliers effectively absorbed the negative supply shock from the Iran-Iraq War.

Core Concept 4.10: Price Slump & Market Restructuring (1980s)

Following the unprecedented price peaks of the 1970s, the 1980s were characterized by a **prolonged slump in global oil prices**, fundamentally driven by a structural realignment of both demand and supply.

(a) **Demand Side Shifts**

On the demand side, the elevated prices of the previous decade catalyzed widespread energy efficiency measures; industrialized nations implemented **strict vehicle fuel economy standards**, transitioned base-load power generation away from petroleum toward **nuclear and coal alternatives**, and began shifting their economies away from energy-intensive heavy manufacturing.

(b) **Supply Side Consequences**

Simultaneously, on the supply side, the artificially high prices maintained by OPEC heavily **incentivized capital-intensive exploration outside the cartel's control**. This resulted in a **massive influx of new, competitive crude oil** from emerging non-OPEC regions, most notably the technological triumphs in the **North Sea** and the **laskan North Slope**, alongside vast offshore discoveries in the **Cantarell Field** by Mexico and aggressive export maximization by the Soviet Union.

This dual force of contracting global consumption and surging non-OPEC production systematically dismantled OPEC's pricing power, leading to a sustained supply glut that culminated in the dramatic **price collapse of 1986** (*Core Concept 4.11*).

Core Concept 4.11: Price Collapse (1986)

By the mid-1980s, the global oil market was heavily saturated due to contracting demand and surging non-OPEC production. To defend high prices, OPEC relied heavily on Saudi Arabia to act as a **swing producer**, unilaterally cutting its own output to offset overproduction by other cartel members.

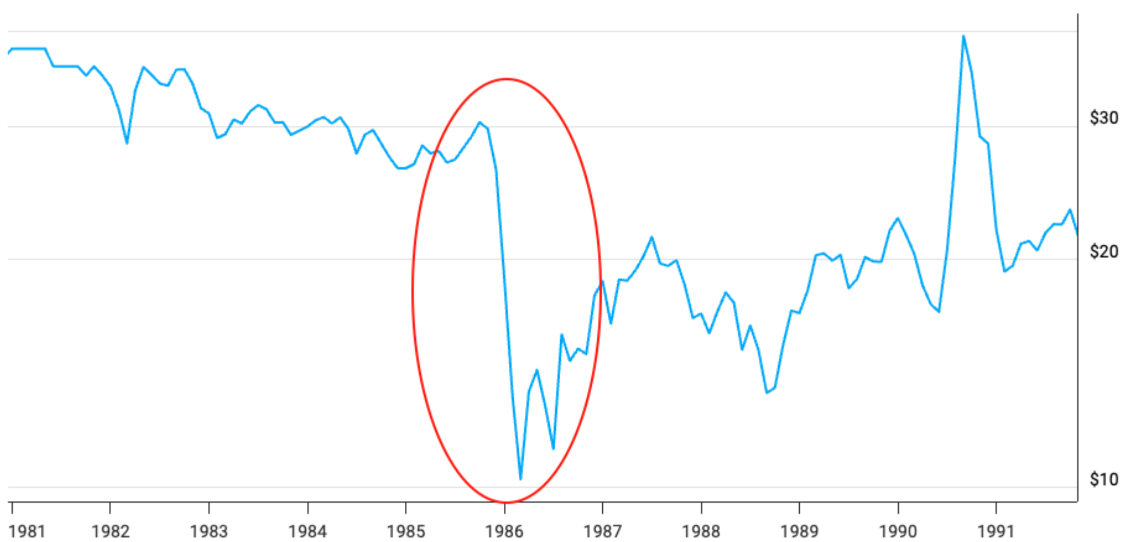

Figure 16: Crude oil prices from Jan 1981 to Sep 1991; *1968 price collapse due to sudden influx of Saudi crude via netback pricing*

However, after watching its market share and revenues plummet, Saudi Arabia fundamentally reversed its strategy in late 1985. Abandoning production quotas, the Kingdom aggressively increased production to reclaim market share and introduced **netback pricing**—a scheme tying crude costs directly to refined product prices, effectively guaranteeing buyer margins and undercutting competitors. This **sudden influx of Saudi crude** overwhelmed an already oversupplied market, causing global oil prices to crash from roughly $27 per barrel to below $10 per barrel in 1986 (*Figure 16*. This historic collapse devastated the economies of oil-exporting nations and decisively **ended the era of artificially sustained high prices**.

4.3.3 The 1990s: Gulf War and Asian Demand Contraction

Core Concept 4.12: Gulf War & Operation Desert Storm (1990)

In August 1990, reeling from massive debts incurred during the Iran-Iraq War, Saddam Hussein ordered the Iraqi military to invade neighboring Kuwait. This prompted the United Nations to impose a **strict trade embargo**, abruptly removing approximately 4.3 million barrels per day of combined crude output from the global market. The subsequent supply disruption **triggered immediate market panic**, causing global oil prices to more than double by October.

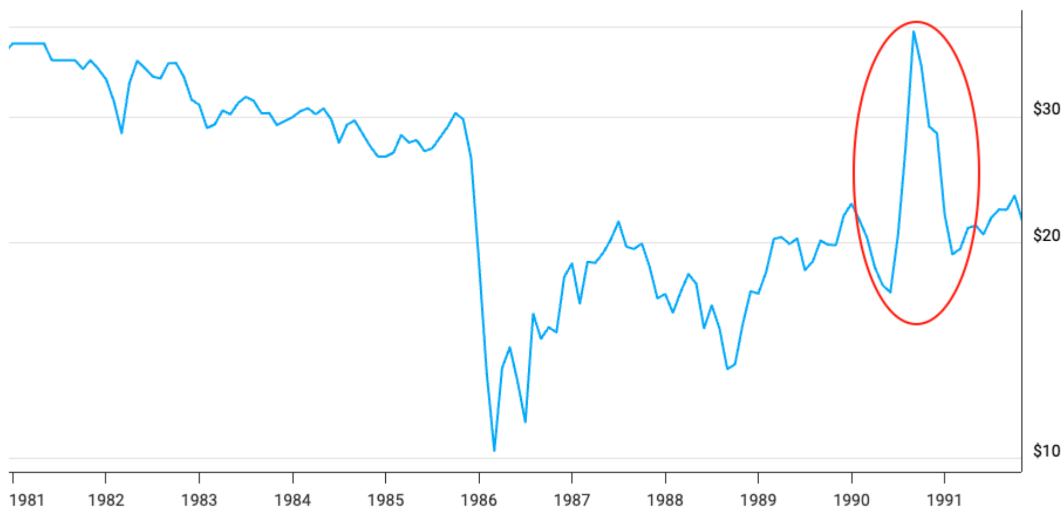

Figure 17: Crude oil prices from Jan 1981 to Sep 1991; *1990 price spike and plummet due to Gulf War and subsequent additional crude production by Saudi Aramco*

However, unlike the prolonged energy crises of the 1970s, this **price spike was remarkably short-lived**. To mitigate the macroeconomic fallout and cover the massive supply deficit, primarily Saudi Arabia's Saudi Aramco (*a member of the New Seven Sisters mentioned in Core Concept 4.4*) rapidly mobilized their spare capacity, significantly surging daily production. This decisive supply intervention, combined with the coordinated release of **strategic petroleum reserves** (*Core Concept 4.6*) by industrialized consumer nations and the swift military resolution of the conflict via **Operation Desert Storm**, successfully calmed the markets and drove oil prices back down to pre-invasion levels by early 1991.

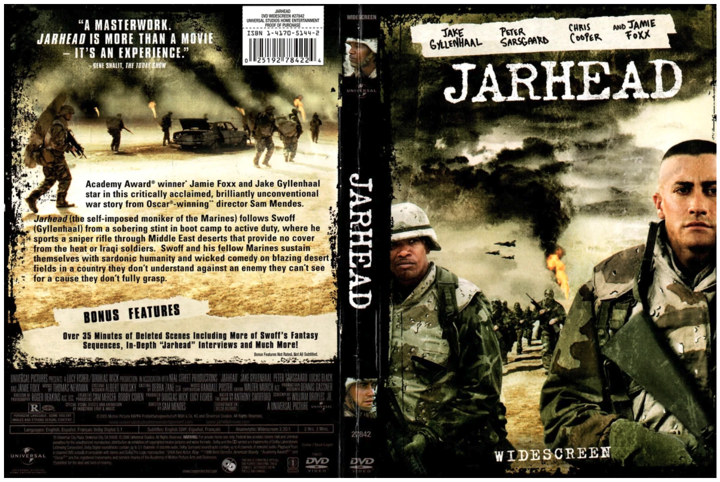

Figure 18: *Jarhead* (2005), a cinematic portrayal of U.S. Marine deployment during the 1990–1991 Gulf War

Core Concept 4.13: Asian Economic Crisis (1997)

The Asian Financial Crisis of 1997 was triggered by the conversion of the Baht to float by the Thailand government, which consequently made foreign investors aggressively pull out money out of the entire Asian market, realizing that Indonesia, South Korea, and Malaysia shared the same structural flaws as Thailand. This severely curtailed economic growth in the region, leading to a sharp decrease in the growth of global oil demand.

Just as the crisis was unfolding in late 1997, OPEC held a ministerial **meeting in Jakarta**. Completely underestimating the severity and global global reach of the Asian crisis, OPEC actually **agreed to increase its production quotas by 10%** to capture what they believed was still a growing market.

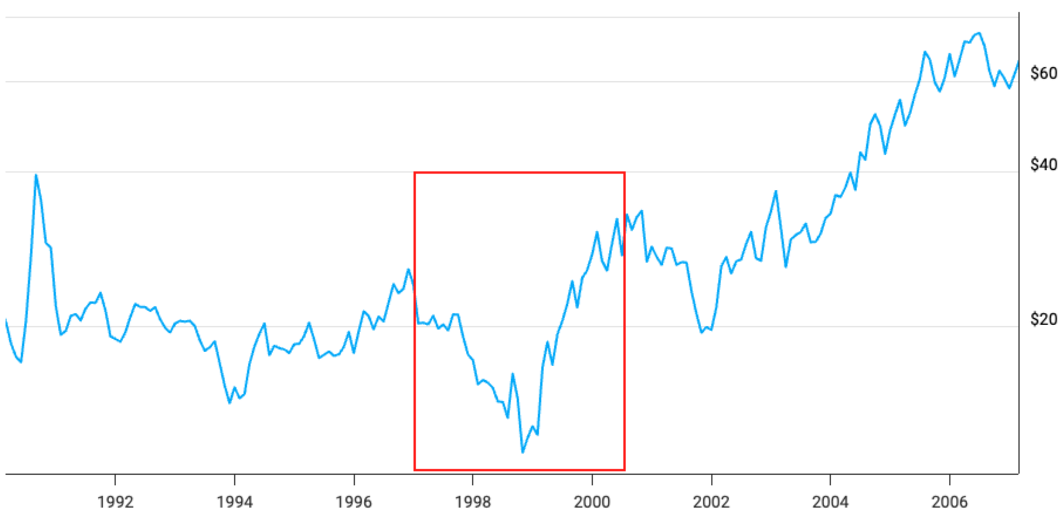

Figure 19: Crude oil prices from Mar 1990 to Feb 2007; *1997 V-shaped price pattern due to the Asian Economic Crisis and subsequent production and IMF corrections*

Following the price collapse to roughly $10 per barrel in late 1998, a swift response led to a market rebound. Recognizing the their quota miscalculation, OPEC executed an emergency reversal in early 1999 by collaborating with major non-OPEC producers to cut global **production cuts by over two million barrels per day**. Concurrently, the affected Asian economies underwent severe but effective **IMF restructuring** programs. These mandated structural banking reforms and massive currency devaluations transformed the affected nations into highly competitive export engines, sparking a rapid V-shaped oil price pattern (*Figure 19*).

4.3.4 The 2000s: The Asian Boom and the Global Financial Crisis

Core Concept 4.14: The Dot-Com Crash, 9/11, and China (Early 2000s)

At the turn of the millennium, global oil markets experienced a brief but severe contraction driven by the bursting of the **Dot-Com bubble** and the profound negative demand shock following the **September 11, 2001 terrorist attacks on the Twin Towers**, which temporarily paralyzed the global aviation sector. While these events triggered a sharp decline in crude prices, the market rapidly stabilized through coordinated supply reductions by OPEC and key non-OPEC producers.

Figure 20: 9/11 terrorist attacks on the Twin Towers in New York City

However, this temporary macroeconomic downturn masked a fundamental structural shift: China officially joining **World Trade Organization (WTO)**—an intergovernmental organization that regulates and facilitates international trade—in late 2001.

Figure 21: Signing ceremony for China's formal accession into the World Trade Organization

China's subsequent industrial expansion provided a vital **demand floor during the Western recession**. As global economies recovered, **export-driven consumption** from China and other emerging markets ignited a massive commodity supercycle (*eventually ended by the global financial crisis in 2008 as covered in Core Concept 4.15*). The ensuing price escalation heavily incentivized a new era of capital-intensive exploration and extraction activities.

Core Concept 4.15: Global Financial Crisis (2008)

The **2008 global financial crisis**, triggered by the bursting of the **U.S. housing bubble**, caused a severe contraction in economic activity and a corresponding collapse in oil demand, driving prices from a record high of $147 per barrel down to roughly $32 per barrel in mere months.

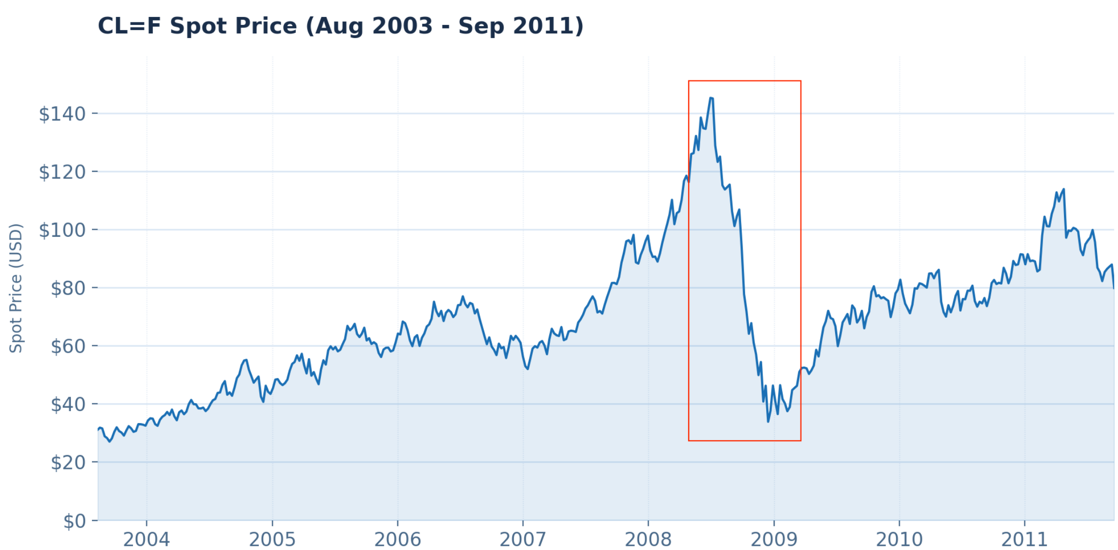

Figure 22: Crude oil future spot from Aug 2003 to Sep 2011; *oil price plummets in mid 2008 due to the financial crisis*

To prevent a prolonged depression, global central banks implemented unprecedented monetary interventions, including **Zero Interest Rate Policies** and **Quantitative Easing**, while nations like China deployed massive fiscal stimulus aimed at heavy infrastructure. Concurrently, **OPEC cut production** to drain excess inventory.

This synchronized influx of global liquidity and aggressive stimulus successfully reignited crude demand and stabilized prices. Furthermore, this era of artificially cheap capital financed the development of hydraulic fracturing, igniting the **US Shale Revolution** (*Core Concept 4.16*. As the global economy recovered, geopolitical fears that the ensuing **Arab Spring** unrest in North Africa (*Core Concept 4.17*) and the Middle East might disrupt major oil-producing nations maintained persistent upward pressure on the rebounding market.

4.3.5 The 2010s: Regional Instability, the US Shale Boom, and Market Restructuring

Core Concept 4.16: Shale Revolution

Following the price recovery subsequent to the 2008 financial crisis, the United States heavily invested in unconventional oil extraction technologies, specifically horizontal drilling and hydraulic fracturing within shale formations. This significant expansion in North American supply, combined with decelerating demand growth in major Asian economies and ongoing geopolitical volatility, precipitated a severe collapse in global oil prices in 2014. OPEC initially responded by lifting supply restrictions in an attempt to drive high-cost US producers out of the market. However, the operational resilience of US shale allowed domestic production to persist, drastically reducing US import reliance and enabling the country to become a net oil exporter in 2018 for the first time in 75 years. This paradigm shift has progressively challenged OPEC's market influence, a dynamic further exacerbated by internal discord and subsequent price wars with non-OPEC producing nations.

Core Concept 4.17: Arab Spring & Regional Instability (2011)

Starting in late 2010, a series of anti-government protests and armed rebellions spread across the Arab world. The civil war in Libya and the subsequent US and NATO intervention in 2011 effectively halted the country's oil exports, removing approximately 1.5 million barrels per day of high-quality sweet crude from the global market. Concurrently, the tightening of economic sanctions on Iran by the US and the European Union in 2012 severely restricted Iranian oil exports. These compounding supply disruptions maintained global oil prices well above $100 per barrel, providing a strong financial incentive for the rapid expansion of US shale production.

Core Concept 4.18: OPEC+ (2016)

In response to the loss of market share caused by the US shale boom, OPEC formed an unprecedented alliance with ten non-OPEC oil-producing nations, most notably Russia, in 2016. This expanded coalition, known as OPEC+, coordinates production quotas to manage global supply and exert greater influence over crude prices.

4.3.6 The 2020s: Pandemics and Geopolitical Conflicts

Core Concept 4.19: COVID-19 Crash and Price War (2020)

In early 2020, a brief but intense price war between Russia and Saudi Arabia led to a severe oversupply, causing oil prices to drop by more than 50 percent. This supply shock was immediately compounded by the COVID-19 pandemic, which triggered an unprecedented collapse in global demand. At the height of the crisis, Brent crude prices plummeted to below $20 per barrel, reaching levels not seen in two decades.

Core Concept 4.20: Russia-Ukraine War (2022)

Russia's invasion of Ukraine in 2022 generated massive uncertainty in global energy markets, sending oil prices to a peak of $127 per barrel. Brent crude remained elevated above $100 throughout the first half of the year as Western nations imposed strict sanctions, and European Union member states actively worked to rapidly phase out their reliance on Russian oil and natural gas imports.

Core Concept 4.21: Strait of Hormuz Crisis (2026)

In early 2026, Iran escalated regional tensions by closing the Strait of Hormuz, a critical global shipping lane for crude oil. This severe disruption to physical market supply caused panic among market participants, leading to a near 100 percent increase in oil prices by the end of April.

5 Metal Markets

6 Agricultural Markets